

The paper uses general theory for **continuous-time arbitrage-free price processes** to develop a framework for the direct modelling and forecasting of realised volatility and correlation.

Using data on the exchange market to estimate a **long-memory Gaussian VAR** for a set of daily logarithmic realized volatilities, the model forecast's volatility at higher rates than conventional GARCH models.

The volatility model generates **well-calibrated density forecasts** in its **quantile** for the *multivariate foreign exchange application*.

The **realised volatility** measure does not distinguish between variability originating from continuous price movements or jumps; hence, improvements are possible by modelling the **jump component**. T

Lognormal-normal work well in the present context, the predictive distribution could be refined and adapted to challenging environments using **numerical simulation methods**. Helping to deal with *conditionally asymmetric return distributions over longer horizons arising from a significant correlation between volatility and return innovations*.

Density forecasts are used to obtain **VaR estimates (quantiles)**, *the same density forecasts can be used to calculate other interests in financial risk management*.

Exchange rate density forecasting could be extended to other classes of financial assets such as U.S equities, portfolio allocation (Busse (1999) and Fleming, Kirby, and Ostdiek (2001, 2002)), applications should be fruitful in high-dimensional settings.

Realised volatility forecasting models that are **parametrized** in ways that guarantee positive definiteness of forecasted covariance matrices within high-dimensional settings, is a key research area.

The in-sample **realised covariance matrix** is *positive definite under quite general conditions one approach would be to model the Cholesky factors rather than the realized covariance matrix itself*.

High-dimensional situations, that allow for **factor structure** in the modelling and forecasting of realised volatility can be useful and is central to both empirical and theoretical financial economics.

Previous research on factor volatility models focused more on *complex procedures involving a latent volatility factor*, while factor analysis of realised volatility is more straight forward.

Identification of explicit volatility factors and its associated market-wide variables that underlie the systematic volatility movements, may help to better understand the 'economics of volatility'

Joint distributional characteristics of asset returns are pivotal, key to pricing instruments relevant to the **risk-return trade-off**,

Conditional portfolios return distributions that govern the likelihood of shifts in portfolio return distributions, govern the shifts in portfolio value, related to risk management.

A critical feature of the **Conditional return distribution** is its **second moment's structure**, forecasting performance has improved with more data, as high-frequency volatility is highly predictable, the information in high-frequency data is useful for forecasting at longer horizons, at monthly or quarterly horizons.

Standard practice of using *daily returns to construct monthly realised variance* results from how the models used to forecast at the daily level cannot readily accommodate the information in intraday data.

Current volatility models consider returns at low dimensionality, such as the multivariate ARCH suffer from the **curse of dimensionality**, constraining their practical application. Consequently, financial professionals use ad hoc methods (feasibility, simplicity and speed of implementation) like exponential smoothing, assuming conditionally normally distributed returns.

Novel improvements to forecast volatility, a VaR calculation the exploits the information in intraday return data, without having to explicitly model the intraday data that has ease of implementation for high-dimensional return volatility modelling.

The **Theory of Quadratic Variation** states that under suitable conditions, realised volatility is an unbiased and highly efficient estimator of return volatility. Their model estimates the multivariate VaR model for the logarithmic realised volatilities, permitting measurement errors within the realised volatilities. The authors VaR produce superior forecasts